Euclidean Dirac upper bound for mean-convex star-shaped hypersurfaces (Theorem 1.1)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2023-029-total-mean-curvature-and-first-dirac-eigenva/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span is the author Theorem environment NewUpperBound with its own label and its own single proof environment (source lines 224-258).

Editorial note. The quasi-spherical display (QS-metric) and the mass inequality (GeneralInequality) sit before the statement; the Herzlich theorem (HerzlichPMTn, printed 2.1) and the corner remark (SpinBoundaryCorner, printed 2.4) are theorem-like contexts placed after the statement so the card keeps 1.1 (both print locally renumbered).

Editorial note. The bibitems EichmairMiaoWang, Herzlich1, Herzlich2 and BartnikChrusciel are extracted verbatim from the pinned source's own thebibliography.

Editorial note. Printed number anchored to the published PDF (SIGMA 19 (2023) 029, read from the stream-extracted text of sigma23-029.pdf downloaded from sigma-journal.com): the rendered labels read Theorem 1.1., Theorem 1.7. and Corollary 1.8. directly, and the PDF destination enumeration Theorem.1.1-1.8 / 2.1-2.4 / 3.1-3.2 / 4.1 / 5.1-5.3 confirms the shared sectionwise theorem-family counter. The pinned tex has five \section commands and no \appendix and its section structure matches the published PDF one-to-one (CONTROL 8.8.109 face).

Editorial note. Equation and theorem numbers inside the card are locally renumbered; every cross-reference is a symbolic \eqref/\ref and resolves inside the card.

Editorial scope. Single environment card cut from the SIGMA article doi:10.3842/SIGMA.2023.029 (sigma-2023-029-total-mean-curvature-and-first-dirac-eigenva); statement and proof are the author's own text at source lines 113-119 and 224-258.

Allowed standard prerequisites (editorial).

- Quasi-spherical metric construction (QS-metric). Delivered in context (source lines 197-199): for any positive initial value u_0 on Sigma there

is a positive u making g_u = u^2 drho^2 + gamma_rho scalar flat and asymptotically flat with well-defined ADM mass (Shi-Tam, cited at source line 196).. Load-bearing: the metric g_u the proof builds with u_0 = H_0/(2 lambda_1) in both cases (source lines 225-229 and 253-257).

- Shi-Tam mass inequality (GeneralInequality). Delivered in context (source lines 209-211): integral H_0 >= integral u_0^{-1} H_0 as soon as the ADM mass of (E, g_u) is nonnegative.. Load-bearing: converts mass nonnegativity into the bound (NewUpperBound1) (source line 229).

- Positive mass with compact inner boundary (Herzlich, Theorem 2.1). Delivered in context (source lines 215-221): the ADM mass is nonnegative when lambda_1 >= H/2, and zero mass forces flatness with constant boundary mean curvature.. Load-bearing: nonnegativity at source line 229 and the equality analysis at source lines 229-233 and 257.

- Eichmair-Miao-Wang deformation. Named construction (cited at source lines 235, 239, 245): a star-shaped hypersurface with positive scalar and mean curvatures is deformed to strictly convex parallels Sigma_rho with a scalar-flat quasi-spherical metric g_v on Sigma x [0, rho_0].. Load-bearing: reduces the star-shaped case to the strictly convex one (source lines 235-257).

- Positive mass across a Lipschitz corner (SpinBoundaryCorner, Remark 2.4). Delivered in context (source lines 285-297): the positive mass theorem holds for asymptotically flat manifolds with corners along a hypersurface.. Load-bearing: extends the Herzlich theorem to the glued manifold with Lipschitz metric (source line 257).

$$g_u := u^2 \, \mathrm{d}\rho^2 + \gamma_\rho \tag{1.1}$$

$$\int_\Sigma \mathcal{H}_0 \, \mathrm{d}\Sigma \geq \int_\Sigma u_0^{-1} \mathcal{H}_0 \, \mathrm{d}\Sigma. \tag{1.2}$$

Theorem 1.1. *Let Σ be a compact, orientable, mean-convex and star-shaped hypersurface with positive scalar curvature in the n-dimensional Euclidean space. Then the first nonnegative eigenvalue $\lambda_1(\Sigma, \gamma)$ of the extrinsic Dirac operator $\not{D}_\gamma$ of (Σ, γ) satisfies*

$$\lambda_1(\Sigma, \gamma) \leq \frac{\int_\Sigma \mathcal{H}_0 \, \mathrm{d}\Sigma}{2|\Sigma|}. \tag{1.3}$$

Here $\mathcal{H}_0$ denotes the mean curvature function of Σ in $\mathbb{R}^n$ and $|\Sigma|$ its volume. Equality occurs if and only if the hypersurface is a round sphere.

Theorem 1.2 ([2, 3])**.** *Let (M, g) be a n-dimensional Riemannian spin asymptotically flat manifold with nonnegative scalar curvature and with a compact inner boundary Σ. Assume that the first nonnegative eigenvalue $\lambda_1(\Sigma, \gamma)$ is positive and satisfies*

$$\lambda_1(\Sigma, \gamma) \geq \frac{1}{2} H, \tag{1.4}$$

where H is the mean curvature of Σ in M. Then the mass is nonnegative and if the mass is zero, (M, g) is flat and the mean curvature is constant and equal to $H = 2\lambda_1(\Sigma, \gamma)$.

Remark 1.3. We briefly explain here how to prove Theorem 1.2 for asymptotically flat manifolds with corners along a hypersurface. An asymptotically flat manifold (M, g) with boundary Σ is said to have a corner along a hypersurface N if it can be written as the disjoint union of two subsets M_+ and M_-, where (M_+, g_+) is a smooth n-dimensional asymptotically flat manifold with boundary N and (M_-, g_-) is a smooth n-dimensional compact Riemannian manifold with boundary $\Sigma \cup N$ and where $g_\pm = g_{|M_\pm}$. The ADM mass of (M, g) corresponds to the ADM mass of (M_+, g_+). Let also H_+ and H_- the mean curvatures of N in (M_+, g_+) and (M_-, g_-) with respect to the unit normal pointing toward infinity. The positive mass theorem for such manifolds asserts that if the scalar curvature of (M, g) is nonnegative and if $H_- \geq H_+$, then its ADM mass is nonnegative. The proof of the positive mass Theorem 1.2 relies essentially on two important points: the existence of a harmonic spinor field satisfying the Atiyah–Patodi–Singer condition which is asymptotic to a constant spinor, usually called a Witten spinor, and an integral version of the Schrödinger–Lichnerowicz formula. For the first point, it is enough to observe that since the metric g is Lipschitz, the work of Bartnik and Chruściel [4] still applies to construct an adequate spinor Φ. On the other hand, recall that the Schrödinger–Lichnerowicz formula asserts that

$$D^2\psi = \nabla^*\nabla\psi + \frac{R}{4}\psi$$

for all spinor field ψ on M and where D, ∇ and ∇^* are respectively the Dirac operator, the spin Levi-Civita connection and its L^2-formal adjoint on (M, g). Although this formula is no longer valid on (M, g) since the scalar curvature is not defined on N, it remains true on (M_+, g_+) and on (M_-, g_-) separately. We can then integrate this formula on M_- first and then on a domain M_R in M_+ whose boundary is the union of N and a coordinate sphere at infinity with radius R. Adding them together and inserting the aforementioned Witten spinor Φ lead to

$$\begin{aligned}\frac{1}{2}(n-1)\omega_{n-1}m_{\mathrm{ADM}}(M, g) = &\int_M \left(|\nabla\Phi|^2 + \frac{R}{4}|\Phi|^2\right) \mathrm{d}M + \frac{1}{2}\int_N (H_- - H_+)|\Phi|^2 \,\mathrm{d}N \\ &- \int_\Sigma \langle \not{D}_\gamma \Phi + \frac{H}{2}\Phi, \Phi\rangle \,\mathrm{d}\Sigma\end{aligned}$$

as R goes to infinity. The nonnegativity of the mass then follows from the assumptions $R \geq 0$, $H_- \geq H_+$ and (1.4).

Proof of Theorem 1.1. First assumed that (Σ, γ) is a strictly convex hypersurface in $\mathbb{R}^n$. As recalled above, one can solve the quasi-spherical metric problem on $\mathcal{E}$ with the initial value

$$u_0 = \frac{1}{2\lambda_1(\Sigma, \gamma)}\mathcal{H}_0 > 0.$$

Then the metric g_u defined by (1.1) yields, on $\mathcal{E}$, an asymptotically flat metric with zero scalar curvature for which the inner compact boundary is isometric to

(Σ, γ) and has constant mean curvature $H = 2\lambda_1(\Sigma, \gamma)$. It is immediate to see that Theorem 1.2 applies so that we deduce from (1.2) the inequality (1.3). If now we assume that equality occurs, then the ADM mass of $(\mathcal{E}, g_u)$ is zero and the positive mass theorem implies that the metric g_u is flat. From the Gauss formula, one can compute that the Riemann curvature tensor Riem_u of g_u is given by

$$\mathrm{Riem}_u(e_i, e_j, e_i, e_j) = \big(1 - u^{-2}\big)\mathrm{Riem}_\rho(e_i, e_j, e_i, e_j),$$

where Riem_ρ is the Riemann curvature tensor of Σ_ρ and $\{e_i \mid 1 \leq i \leq n-1\}$ is a local orthonormal frame on (Σ, γ) parallel translated in the direction of $\frac{\partial}{\partial \rho}$. Since the metric g_u is flat and Σ_ρ is strictly convex in $\mathbb{R}^n$ for all $\rho \geq 0$, we conclude that $u \equiv 1$. Then (Σ, γ) is a smooth embedded compact hypersurface in the Euclidean space with constant mean curvature and so it is a round sphere. The converse is obvious since for a Euclidean round sphere with radius $r > 0$, its first Dirac eigenvalue is equal to $(n-1)/(2r)$ and its mean curvature is equal to $(n-1)/r$.

If now we assume that the hypersurface Σ is star-shaped with positive scalar and mean curvatures, we apply the method developed by Eichmair, Miao and Wang [1]. Under these assumptions, there exists a smooth map $F\colon \Sigma \times [0, \infty) \to \mathbb{R}^n$ such that

$$\frac{\partial F}{\partial \rho} = \frac{\mathcal{H}_\rho}{\not{R}_\rho}\nu,$$

$F(\Sigma, 0) = \Sigma$ and $\Sigma_\rho := F(\Sigma, \rho)$ has positive mean and scalar curvatures and is a strictly convex hypersurface for ρ sufficiently large, say for $\rho \geq \rho_0$. Here ν is the outer unit normal to the hypersurface Σ_ρ, $\mathcal{H}_\rho$ is its mean curvature and $\not{R}_\rho$ its scalar curvature. Moreover, the pull-back of the Euclidean metric by F on $\Sigma \times [0, \rho_0]$ has the form g_η given by (1.1) with η a smooth positive function. From [1, Proposition 2], there exists a smooth positive function v on $\Sigma \times [0, \rho_0]$ such that the scalar curvature of the metric g_v is zero and with initial data

$$v(\cdot, 0) = \frac{\eta(\cdot, 0)}{2\lambda_1(\Sigma, \gamma)}\mathcal{H}_0.$$

This last condition implies that the mean curvature of $\Sigma \simeq \Sigma \times \{0\}$ is equal to $h_0 = 2\lambda_1(\Sigma, \gamma)$ for the metric g_v. Then, as proved in [1, Proposition 3], the function

$$\rho \in [0, \rho_0] \mapsto \int_{\Sigma_\rho} \big(\mathcal{H}_\rho - h_\rho\big)\, \mathrm{d}\Sigma_\rho,$$

where h_ρ represents the mean curvature of Σ_ρ with respect to the metric g_v, is monotone nonincreasing in ρ and so it follows that

$$\int_\Sigma \big(\mathcal{H}_0 - h_0\big)\, \mathrm{d}\Sigma = \int_\Sigma \mathcal{H}_0\, \mathrm{d}\Sigma - 2\lambda_1(\Sigma, \gamma)|\Sigma| \geq \int_\Sigma (\mathcal{H}_{\rho_0} - h_{\rho_0})\, \mathrm{d}\Sigma. \tag{1.5}$$

Now we can apply the original construction of Shi and Tam to the strictly convex hypersurface Σ_{ρ_0} by solving the quasi-spherical equation in such a way that the resulting manifold $(\Sigma \times [\rho_0, \infty), g_u)$ is an asymptotically flat manifold whose

inner boundary is isometric to $(\Sigma, \gamma_{\rho_0})$ with mean curvature equal to h_{ρ_0}. As before, we get from (1.2) that

$$\int_\Sigma (\mathcal{H}_{\rho_0} - h_{\rho_0})\, \mathrm{d}\Sigma \geq c(n) m_{\mathrm{ADM}}(\mathcal{E}, g_u). \tag{1.6}$$

Gluing $(\Sigma \times [0, \rho_0], g_v)$ and $(\Sigma \times [\rho_0, \infty), g_u)$ along their common boundary $(\Sigma, \gamma_{\rho_0})$ yields an asymptotically flat manifold $(\mathcal{E}, \widetilde{g})$ with Lipschitz metric along Σ_{ρ_0}, with ADM mass equal to $m_{\mathrm{ADM}}(\mathcal{E}, g_u)$ and with a compact inner boundary whose mean curvature is equal to $2\lambda_1(\Sigma, \gamma)$. Combining (1.5) and (1.6) with the fact that Theorem 1.2 holds in this context (see Remark 1.3) yields the desired inequality. If equality occurs, we deduce as before that the manifold $(\mathcal{E}, \widetilde{g})$ is flat away from Σ_{ρ_0} and that $u \equiv 1$ since Σ_ρ is convex for $\rho \geq \rho_0$. On the other hand, we also deduce from the Gauss formula that $u \equiv \eta$ since Σ_ρ has positive scalar curvature for $\rho \in [0, \rho_0]$. This allows to conclude that $(\mathcal{E}, \widetilde{g})$ is isometric to the exterior of Σ in the Euclidean space. The end of the proof then proceeds as in the previous case. □

References

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